1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

In breast cancer patients treated with neoadjuvant therapy, the predictive value of the PAM50 proliferation score for achieving pathologic complete response (pCR) is conditional on the density of tertiary lymphoid structures (TLS) within the tumor microenvironment. Specifically, high PAM50 proliferation coupled with high TLS density predicts a higher likelihood of pCR due to an enhanced local anti-tumor immune response, whereas high proliferation with low TLS density predicts a lower pCR rate. This interaction suggests that TLSs modulate the efficacy of chemotherapy in proliferative tumors by facilitating local antigen presentation and cytotoxic immune activity. Secondary hypotheses include: (a) TLS density provides incremental predictive value beyond PAM50 proliferation, TIL density, and standard clinicopathologic factors; (b) TLS density and TIL density together stratify relapse-free and overall survival beyond proliferation status; (c) among molecular subtypes with higher baseline immunogenicity (e.g., luminal B and HER2-enriched), the proliferation-TLS interaction will be most pronounced. The primary endpoint is pCR (binary) after neoadjuvant therapy; secondary endpoints include relapse-free survival (RFS) and overall survival (OS). Anchor genes to test (subject to Ontologist input) may include immune-related genes such as HLA-DQB1 and other HLA class II genes to reflect antigen presentation within TLS, along with B- and T-cell markers (e.g., CD19/CD20, CD3, CD8) to characterize TLS components.

2. Why This Gene — Not Others

3. Clinically Testable Predictions

Prediction 1: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions